

I solved the questions before the update, so the answers were written according to 0.01.

Use the normal and psoriasis transcriptome data reported in Jabbari et al. (2014) to calculate significance of expression change for all the genes between the two groups. Perform the following steps:

(a) Use GEO2R interface to compare the two groups for a Benjamini-Hochberg corrected p-value cutoff of 0.01. Download the results, and save in a text file.

ncbi.nlm.nih.gov/gds/?term=Jabbari%20Dominant%20Th1%20and%20minimal%20Th17%20skewing

An official website of the United States government [Here's how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

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GEO DataSets

Create alert Advanced Help

Entry type
DataSets (1)
Series (1)
Samples (0)
Platforms (0)

Organism
Customize ...

Study type
Expression profiling by array
Methylation profiling by array
Customize ...

Author
Customize ...

Attribute name
tissue (2)
strain (0)
Customize ...

Summary Sort by Default order

Search results
Items: 2

☐ [Dominant Th1 and Minimal Th17 Skewing in Discoid Lupus Revealed by Transcriptomic Comparison with Psoriasis](#)
(Submitter supplied) Discoid lupus erythematosus (DLE) is the most common skin manifestation of lupus. Despite its high frequency in systemic lupus in addition to cases without extracutaneous manifestations, targeted treatments for DLE are lacking, likely because of a dearth of knowledge of the molecular landscape of DLE skin. Here, we profiled the transcriptome of DLE skin in order to identify signaling pathways and cellular signatures that may be targeted for treatment purposes. [more...](#)
Organism: Homo sapiens
Type: Expression profiling by array
Dataset: GDS4891 Platform: GPL571 38 Samples
Download data: CEL
Series Accession: GSF52471 ID: 200052471

Send to: **Filters:** [Manage Filters](#)

Top Organisms [Tree](#)
Homo sapiens (2)

Find related data
Database:

Search details
Jabbari[All Fields] AND Dominant[All Fields] AND Th1[All Fields] AND minimal[All Fields] AND Th17[All Fields] AND skewing[All Fields]

After clicking on the link in the figure, we proceed with the GEO2R analysis.

ncbi.nlm.nih.gov/geo/geo2r/?acc=GSE52471

NCBI GEO Publications FAQ MIAME Email GEO

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Use GEO2R to compare two or more groups of Samples in order to identify genes that are differentially expressed across experimental conditions. Results are presented as a table of genes ordered by significance. [Full instructions](#) [YouTube](#)

GEO accession [Dominant Th1 and Minimal Th17 Skewing in Discoid Lupus Revealed by Transcriptomic Comparison with Psoriasis](#)

Samples [Define groups](#) Selected 0 out of 38 samples

Group	Accession	Title	Source name	Condition	Patient	Tissue	Kit
-	GSM1267412	Psoriasis-Patient-A	human skin biopsy	Psoriasis	PS-A	LS skin biopsy	Affy
-	GSM1267413	Psoriasis-Patient-B	human skin biopsy	Psoriasis	PS-B	LS skin biopsy	Affy
-	GSM1267414	Psoriasis-Patient-C	human skin biopsy	Psoriasis	PS-C	LS skin biopsy	Affy
-	GSM1267415	Psoriasis-Patient-D	human skin biopsy	Psoriasis	PS-D	LS skin biopsy	Affy
-	GSM1267416	Psoriasis-Patient-E	human skin biopsy	Psoriasis	PS-E	LS skin biopsy	Affy
-	GSM1267417	Psoriasis-Patient-F	human skin biopsy	Psoriasis	PS-F	LS skin biopsy	Affy
-	GSM1267418	Psoriasis-Patient-G	human skin biopsy	Psoriasis	PS-G	LS skin biopsy	Affy
-	GSM1267419	Psoriasis-Patient-H	human skin biopsy	Psoriasis	PS-H	LS skin biopsy	Affy
-	GSM1267420	Psoriasis-Patient-I	human skin biopsy	Psoriasis	PS-I	LS skin biopsy	Affy
-	GSM1267421	Psoriasis-Patient-J	human skin biopsy	Psoriasis	PS-J	LS skin biopsy	Affy
-	GSM1267422	Psoriasis-Patient-K	human skin biopsy	Psoriasis	PS-K	LS skin biopsy	Affy

I sorted the data according to conditions; there are three different conditions: psoriasis, normal, and discoid lupus. Since we are asked for psoriasis and normal, I will divide them into two groups: Pso and Normal.

ncbi.nlm.nih.gov/geo/geo2r/?acc=GSE52471

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Use GEO2R to compare two or more groups of Samples in order to identify genes that are differentially expressed across experimental conditions. Results are presented as a table of genes ordered by significance. Full Instructions [Yes](#) [No](#)

GEO accession Set Dominant Th1 and Minimal Th17 Skewing in Discoid Lupus Revealed by Transcriptomic Comparison with Psoriasis

Selected 31 out of 38 samples

Sample	Accession	Group	Condition	Platform	Assay	
Pso	GSM1267420	human skin biopsy	Psoriasis	PS-I	LS skin biopsy Affy	
Pso	GSM1267421	human skin biopsy	Psoriasis	PS-J	LS skin biopsy Affy	
Pso	GSM1267422	human skin biopsy	Psoriasis	PS-K	LS skin biopsy Affy	
Pso	GSM1267423	human skin biopsy	Psoriasis	PS-L	LS skin biopsy Affy	
Pso	GSM1267424	Psoriasis-Patient-M	human skin biopsy	Psoriasis	PS-M	LS skin biopsy Affy
Pso	GSM1267425	Psoriasis-Patient-N	human skin biopsy	Psoriasis	PS-N	LS skin biopsy Affy
Pso	GSM1267426	Psoriasis-Patient-O	human skin biopsy	Psoriasis	PS-O	LS skin biopsy Affy
Pso	GSM1267447	Psoriasis-252	human skin biopsy	252	LS skin biopsy Ovation	
Pso	GSM1267448	Psoriasis-287	human skin biopsy	287	LS skin biopsy Ovation	
Pso	GSM1267449	Psoriasis-292	human skin biopsy	292	LS skin biopsy Ovation	
Normal	GSM1267427	Normal-N_LJ_1	human skin biopsy	Normal	N_LJ_1	Normal skin biopsy Affy
Normal	GSM1267428	Normal-N_LJ_2	human skin biopsy	Normal	N_LJ_2	Normal skin biopsy Affy
Normal	GSM1267429	Normal-N_LJ_3	human skin biopsy	Normal	N_LJ_3	Normal skin biopsy Affy

Since Benjamin & Hochberg was already selected in the Options section, it was analyzed without modification. Only the value, which was currently 0.05, was changed to 0.01.

GEO2R Options Profile graph R script

Apply adjustment to the P-values. [More...](#)

☒ Benjamini & Hochberg (False discovery rate)

☐ Benjamini & Yekutieli

☐ Bonferroni

☐ Holm

Apply log transformation to the data. [More...](#)

☒ Auto-detect

☐ Yes

☐ No

Apply limma precision weights (vooma). [More...](#)

☐ Yes

☒ No

Force normalization. [More...](#)

☐ Yes

☒ No

Category of Platform annotation to display on results.

☐ Submitter supplied

☒ NCBI generated

Plot displays. [More...](#)

Significance level cut-off (enter number between 0 and 1)

Log 2 fold change threshold

Volcano and Mean-difference plot contrasts (select up to 5)

0 selected (clear)

☐ Pso vs Normal

If you edit Options after performing an analysis, click [Reanalyze](#) to apply the edits:

[Reanalyze](#)

Top differentially expressed genes [?]

[Download full table](#)

[Select columns](#)

Download full table seçeneği ile indirilip text dosyası halinde saklanmıştır. TSV formatında indirilmiştir ama TXT şekilde açılabilir bir formattır.

GEO accession Set Dominant Th1 and Minimal Th17 Skewing in Discoid Lupus Revealed by Transcriptomic Comparison with Psoriasis

I learned the GEO accession code from here.

(b) Get the same dataset via GEOquery package in R based on what you learned in the class. (Hint: Search the GEO to find the dataset). Select 31 samples mentioned above.

```
> b$title
[1] "Psoriasis-Patient-A" "Psoriasis-Patient-B" "Psoriasis-Patient-C"
[4] "Psoriasis-Patient-D" "Psoriasis-Patient-E" "Psoriasis-Patient-F"
[7] "Psoriasis-Patient-G" "Psoriasis-Patient-H" "Psoriasis-Patient-I"
[10] "Psoriasis-Patient-J" "Psoriasis-Patient-K" "Psoriasis-Patient-L"
[13] "Psoriasis-Patient-M" "Psoriasis-Patient-N" "Psoriasis-Patient-O"
[16] "Normal-N_LJ_1"      "Normal-N_LJ_2"      "Normal-N_LJ_3"
[19] "Normal-N_LJ_4"      "Normal-N_LJ_5"      "Normal-N_LJ_6"
[22] "Normal-N_LJ_7"      "Normal-N_LJ_8"      "Normal-N_LJ_9"
[25] "Normal-N_LJ_10"     "Discoid Lupus-L1"    "Discoid Lupus-L2"
[28] "Discoid Lupus-L3"    "Discoid Lupus-L4"    "Discoid Lupus-L5"
[31] "Discoid Lupus-L8"    "Discoid Lupus-L9"    "Normal-N_AJ1"
[34] "Normal-N_AJ2"        "Normal-N_AJ3"        "Psoriasis-252"
[37] "Psoriasis-287"       "Psoriasis-292"

> l
) pval = NULL
for (i in 1:17440) {
  pval[i] = t.test(a[i ,c(1:15, 36:38)],a[i ,c(16:25, 33:35)])$p.value
}
sum(pval <= 0.01)
```

I learned the order of the b\$title users I wanted to select, and then for each gene (17,440), the t-test was applied individually using a for loop, and the p-values were recorded.

(c) Apply t-test to the data to compare normal and psoriasis samples. What is the number of significantly differentially expressed genes for a cut-off of 0.05? What is it for a cut-off of 0.01? What is it for a cut-off of 0.05 if Bonferroni correction is applied?

```
sum(pval <= 0.01)
sum(pval <= 0.05)

> sum(pval <= 0.01)
[1] 6838
> sum(pval <= 0.05)
[1] 9119
```

Using the sum function, we determined how many meet the desired conditions by producing True for those that meet them and false for those that do not (1,0), then adding these together and returning the result. The number of p values less than or equal to **0.01 is 6838**. The number of p values less than or equal to **0.05 is 9119**.

```
bonferroni_pval = NULL
num = 0
for (i in pval) {
  num = num + 1
  bonferroni_p = (i*17440)
  bonferroni_pval[num] = bonferroni_p
}
sum(bonferroni_pval<= 0.05)

> sum(bonferroni_pval<= 0.05)
[1] 1678
```

Each gene's p-value was multiplied by the total n value, and the new p-values were recorded. (Some will be greater than one, which is statistically impossible, but since we are not concerned with that, I did not address it.)

The sum function was used again to count the number of p values less than or equal to **0.05, and 1678** differentially expressed genes were found

(d) Apply Benjamini-Hochberg (false discovery rate) correction using p.adjust function for the same comparison for a p-value cut-off of 0.01. (Hint: the usage is p.adjust (p, method = "BH"), where the input p is the vector of not-corrected p-values).

```

7
} p_adjusted_bh = p.adjust(pval, method = "BH")
} sum(p_adjusted_bh <= 0.01)
}

```

```

> p_adjusted_bh = p.adjust(pval, method = "BH")
> sum(p_adjusted_bh <= 0.01)
[1] 5660

```

The Benjamini-Hochberg method was applied using the `p.adjust` function as requested in the question, with `method = "BH,"` and 5660 differentially expressed genes were found using the `sum` function.

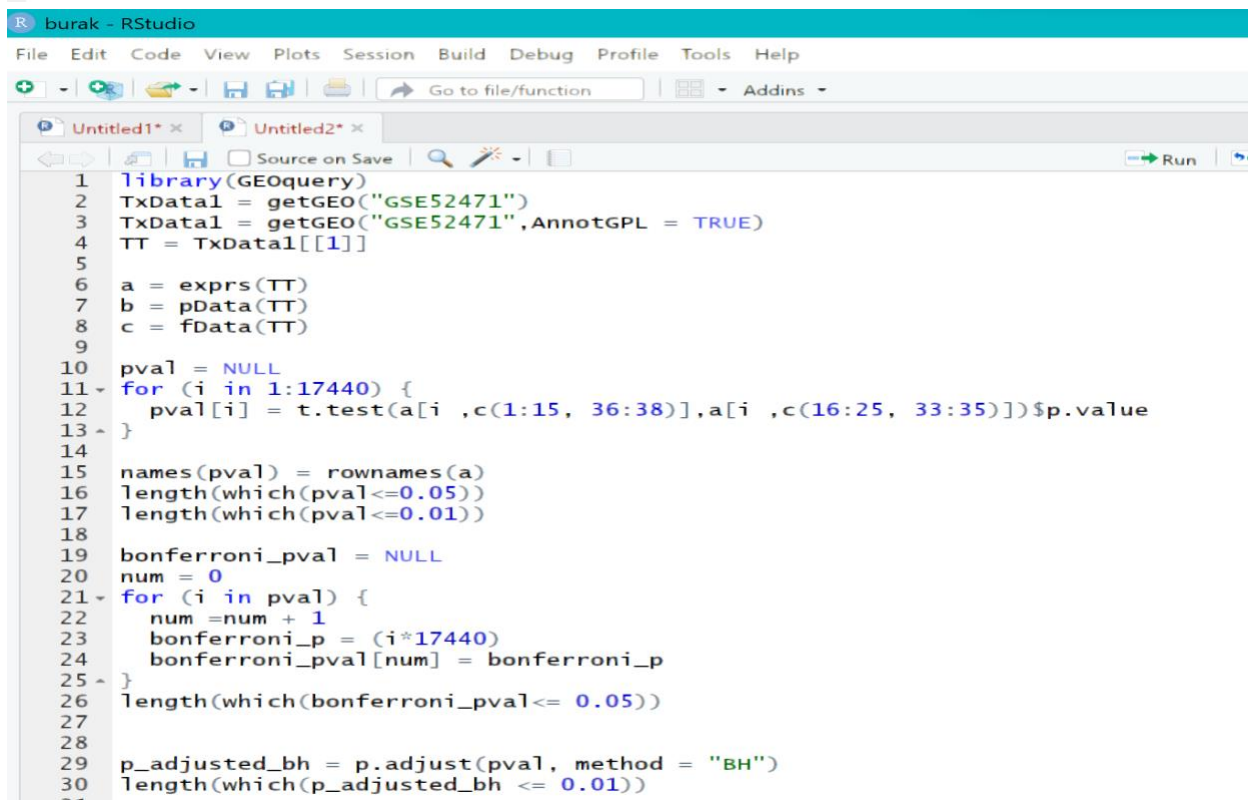
(e) Compare your results with the results you obtained in (a) from the GEO2R interface with the ones you obtained in (d) with R's `p.adjust` function. Did you get the same results? (Hint: you can compare the two gene lists using R's `intersect()` function to find the number of common genes)

When we open the `GSE52471.top.table.tsv` file and look at it, we see that it gives us the p-values for all genes, not just the significant genes according to our analysis. Therefore, we need to add the cutoff ourselves and analyze it again using R. Regarding this question, since I didn't save the gene names and only used the `sum` function to find the number, I decided to save the number and names using `length(which())` and `names(pval) = rownames(a)` instead, and I made the change.

```

names(pval) = rownames(a)
length(which(pval<=0.05))
length(which(pval<=0.01))

```

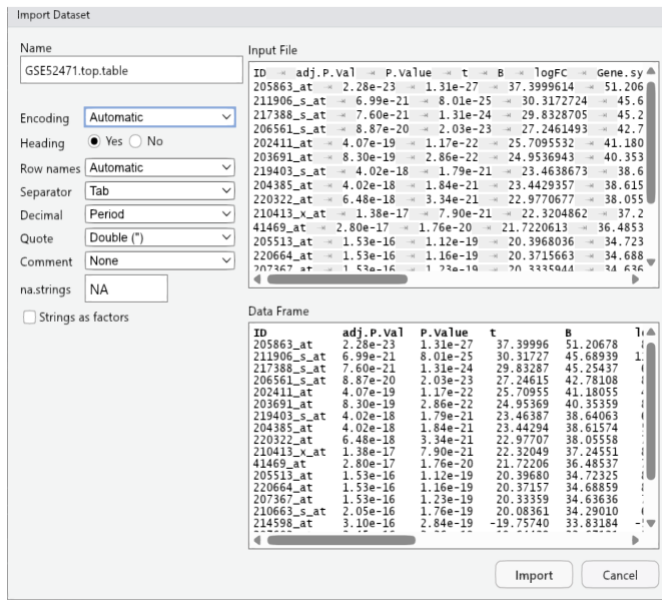


```

burak - RStudio
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Addins
Run

1 library(GEOquery)
2 TxData1 = getGEO("GSE52471")
3 TxData1 = getGEO("GSE52471", AnnotGPL = TRUE)
4 TT = TxData1[[1]]
5
6 a = exprs(TT)
7 b = pData(TT)
8 c = fData(TT)
9
10 pval = NULL
11 for (i in 1:17440) {
12   pval[i] = t.test(a[i, c(1:15, 36:38)], a[i, c(16:25, 33:35)])$p.value
13 }
14
15 names(pval) = rownames(a)
16 length(which(pval<=0.05))
17 length(which(pval<=0.01))
18
19 bonferroni_pval = NULL
20 num = 0
21 for (i in pval) {
22   num = num + 1
23   bonferroni_p = (i*17440)
24   bonferroni_pval[num] = bonferroni_p
25 }
26 length(which(bonferroni_pval<= 0.05))
27
28
29 p_adjusted_bh = p.adjust(pval, method = "BH")
30 length(which(p_adjusted_bh <= 0.01))
31

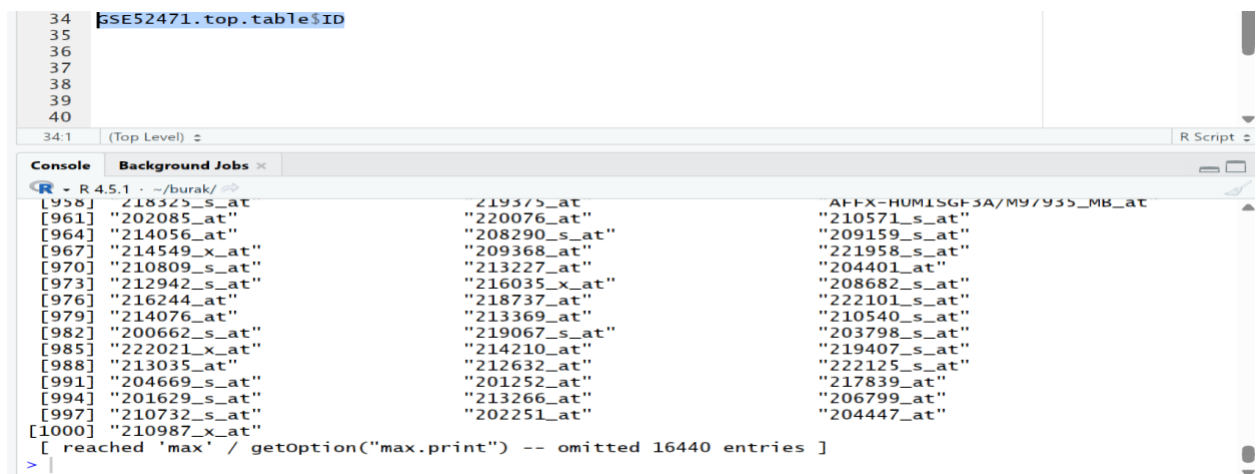
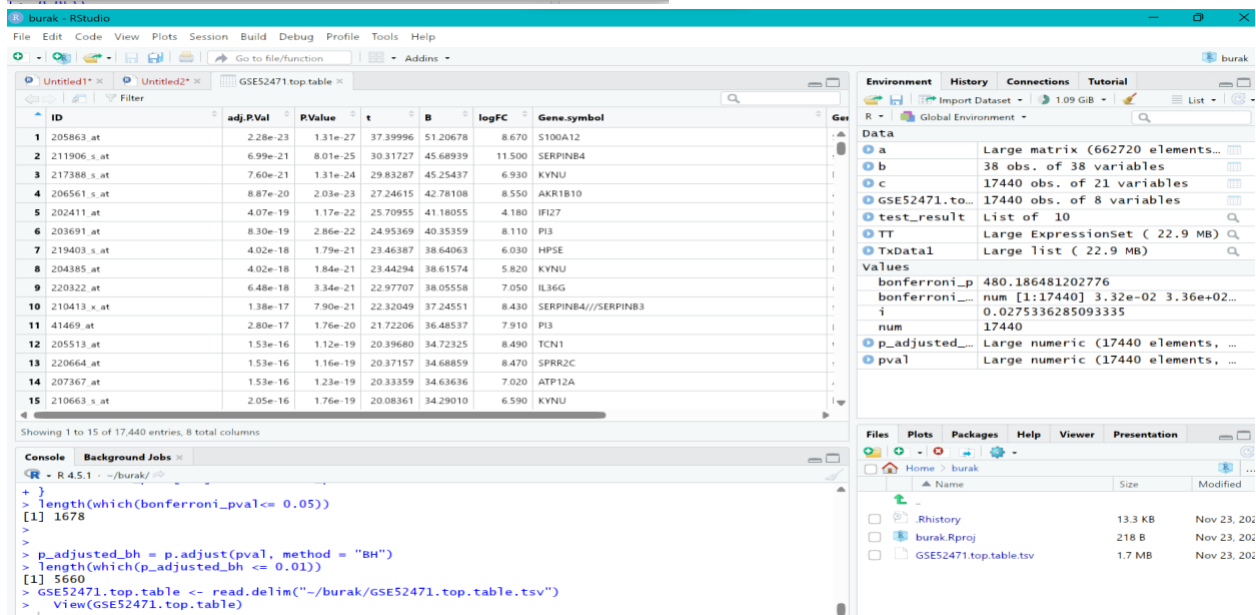
```



I imported the file I downloaded using Import dataset into R. Since the file I obtained with GEO2R provided the p-values for all genes rather than just those at the cutoff we specified, I redefined the cutoff in R, filtered the data, and then used the intersect() function to find the number of genes that were the same.

The table shows that 7,107 differentially expressed genes were found when the p-value cutoff was 0.01 or less.

I found the number of common differentially expressed genes to be 5,646.




```

p_adjusted_bh = p.adjust(pval, method = "BH")
length(which(p_adjusted_bh <= 0.01))
genes = names(which(p_adjusted_bh <= 0.01))

genesGEO = GSE52471.top.table$ID[GSE52471.top.table$P.Value <= 0.01]
length(genesGEO)

same_genes = intersect(genes,genesGEO)
length(same_genes)

```

```

> p_adjusted_bh = p.adjust(pval, method = "BH")
> length(which(p_adjusted_bh <= 0.01))
[1] 5660
> genes = names(which(p_adjusted_bh <= 0.01))
>
>
> genesGEO = GSE52471.top.table$ID[GSE52471.top.table$P.Value <= 0.01]
> length(genesGEO)
[1] 7107
>
> same_genes = intersect(genes,genesGEO)
> length(same_genes)
[1] 5646
>

```

```

40
41 genesGEO = GSE52471.top.table$ID[GSE52471.top.table$adj.P.Val <= 0.01]
42 length(genesGEO)
43
44 same_genes = intersect(genes,genesGEO)
45 length(same_genes)
46
47
48
49
50
51

```

41:1 (Top Level) ↕

Console Background Jobs ✕

R R 4.5.1 · ~/burak/ ↗

```

> View(GSE52471.top.table)
> genesGEO = GSE52471.top.table$ID[GSE52471.top.table$adj.P.Val <= 0.01]
> length(genesGEO)
[1] 6051
>
> same_genes = intersect(genes,genesGEO)
> length(same_genes)
[1] 5506
>

```

The code and outputs for the steps I mentioned on the previous page are provided on the left.

The p-values of the genes in the table were used for comparison. Since it was not specified in the question, I used the first one. Since the adjusted p-value is actually more reliable, I used it in the next image and provided the results.

When the cutoff is set to **0.01** with the adjusted p-value, **6051** dfexprs genes are found.

The common dfexprs gene count for both is 5506. We found a value close to the non-logarithmic form, which actually indicates that the non-logarithmic form is close to a normal distribution.

burak - RStudio

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Go to file/function Addins

Untitled1* x Untitled2* x

Source on Save Run

```

50
51 length(which(pval<=0.05))
52 length(which(pval<=0.01))
53
54 bonferroni_pval = NULL
55 num = 0
56 for (i in pval) {
57   num = num + 1
58   bonferroni_p = (i*17440)
59   bonferroni_pval[num] = bonferroni_p
60 }
61 length(which(bonferroni_pval<= 0.05))
62
63
64 p_adjusted_bh = p.adjust(pval, method = "BH")
65 length(which(p_adjusted_bh <= 0.01))
66 genes = names(which(p_adjusted_bh <= 0.01))
67
68 |
69 genesGEO = GSE52471.top.table$ID[GSE52471.top.table$adj.P.Val <= 0.01]
70
71 same_genes = intersect(genes,genesGEO)
72 cleaned_same_genes = sub("/.*","",same_genes)
73 write.table(cleaned_same_genes,file = "same_gene_sonuc.txt",row.names = FALSE)
74
75
> length(cleaned_same_genes)
[1] 4409
>

```

Untitled1* x Untitled2* x

Source on Save Run

```

61 length(which(p_adjusted_bh <= 0.01))
62 genes = names(which(p_adjusted_bh <= 0.01))
63
64
65 genesGEO = GSE52471.top.table$Gene.symbol[GSE52471.top.table$adj.P.Val <= 0.01]
66
67 same_genes = intersect(genes,genesGEO)
68 cleaned_same_genes = sub("/.*","",same_genes)
69 write.table(cleaned_same_genes,file = "same_gene_sonuc.txt",row.names = FALSE)
70 length(cleaned_same_genes)
71
72
73
74 a = exprs(TT)
75 a = log2(a)
76 pval = NULL
77 for (i in 1:17440) {
78   pval[i] = t.test(a[i ,group1_cols],a[i ,group2_cols])$p.value
79 }
80 names(pval) = rownames(a)
81 p_adjusted_bh = p.adjust(pval, method = "BH")
82 length(which(p_adjusted_bh <= 0.01))
83 genes = names(which(p_adjusted_bh <= 0.01))
84
85 cleaned_genes = sub("/.*","",genes)
86 write.table(cleaned_genes,file = "loged_gene_result.txt",row.names = FALSE)
87
88
67:1 (Top Level)

```

Console Background Jobs

R 4.5.1 ~ /burak/

```

> same_genes = intersect(genes,genesGEO)
> cleaned_same_genes = sub("/.*","",same_genes)
> write.table(cleaned_same_genes,file = "same_gene_sonuc.txt",row.names = FALSE)
> same_genes = intersect(genes,genesGEO)
> cleaned_same_genes = sub("/.*","",same_genes)
> write.table(cleaned_same_genes,file = "same_gene_sonuc.txt",row.names = FALSE)
> length(cleaned_same_genes)
[1] 4409
>

```

(f) Repeat (d) by first bringing the expression data closer to normal distribution by applying logarithm transformation. What is the number of differentially expressed genes this time with the same Benjamini-Hochberg correction?

```

52
53 a = exprs(TT)
54 a = log2(a)
55 pval = NULL
56 for (i in 1:17440) {
57   pval[i] = t.test(a[i ,c(1:15, 36:38)],a[i ,c(16:25, 33:35)])$p.value
58 }
59
60 p_adjusted_bh = p.adjust(pval, method = "BH")
61 length(which(p_adjusted_bh <= 0.01))
62 genes = names(which(p_adjusted_bh <= 0.01))
63
64
65
66
67
68
69

```

Expression levels were converted to log2 and reanalyzed, this time found **5621** dfexprs genes.

```

Console Background Jobs x
R - R 4.5.1 - ~/burak/
> a = exprs(TT)
> a = log2(a)
> pval = NULL
> for (i in 1:17440) {
+   pval[i] = t.test(a[i ,c(1:15, 36:38)],a[i ,c(16:25, 33:35)])$p.value
+ }
>
> p_adjusted_bh = p.adjust(pval, method = "BH")
> length(which(p_adjusted_bh <= 0.01))
[1] 5621
> genes = names(which(p_adjusted_bh <= 0.01))
>

```

I am sharing the code below, which I later learned about online and then corrected. There are no errors in the previous images; this is just a more professional and better-written version. It has been corrected in the R code file.

```

group1_cols = grep("^Psoriasis", b$title)
group2_cols = grep("^Normal", b$title)

```

```

R - burak - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins
Untitled1* x Untitled2* x b x
Source on Save Run
1 library(GEOquery)
2 TxData1 = getGEO("GSE52471")
3 TxData1 = getGEO("GSE52471",AnnotGPL = TRUE)
4 TT = TxData1[[1]]
5
6 a = exprs(TT)
7 b = pData(TT)
8 c = fData(TT)
9
10 group1_cols = grep("Psoriasis", b$title)
11 group2_cols = grep("Normal", b$title)
12
13
14 pval = NULL
15 for (i in 1:17440) {
16   pval[i] = t.test(a[i ,group1_cols],a[i ,group2_cols])$p.value
17 }
18
19 names(pval) = rownames(a)
20 length(which(pval<=0.05))
21 length(which(pval<=0.01))
22
23 bonferroni_pval = NULL
24 num = 0
25

```


(g) Use g:Profiler tool (<https://biit.cs.ut.ee/gprofiler>) to perform Gene Ontology and pathway enrichment analysis of the differentially expressed genes for Benjamini-Hochberg-corrected lists obtained in (d) and (f). Use default options/cut-offs in the online tool.

When I tried it many many times I got a parse error from the website. I realized that the reason was that the website gave an error because some lines contained multiple similar genes, or because they were on the same line with this symbol. I found a function online to fix this, used it to correct the genes, and submitted them to the site. It still gave the same error; no matter what I tried, I couldn't fix it, and I didn't understand why. I think there's an issue with the website because I tried it a few times the previous day, and it worked, but this time I couldn't get it to work at all. For this reason, I used a different website.

ShinyGO 0.85.1
Human

55616_at
55662_at
55692_at
55705_at
55872_at
56197_at

5605 IDs mapped to 2794 (50%) human genes.
Background (recommended) Submit

Pathway database: KEGG

FDR cutoff: 0,05 # pathways to show: 20

Pathway size: Min. 2 Max. 5000

Enrichment FDR	nGenes	Pathway Genes	Fold Enrichment	Pathways (click for details)
3.2E-11	25	46	4.9	Proteasome
1.1E-12	36	80	4.1	Drug metabolism-other enzymes
1.3E-09	29	69	3.8	Chemical carcinogenesis-DNA adducts
1.9E-08	38	120	2.9	Thyroid hormone signalling pathway
3.1E-10	49	158	2.8	Cell cycle
1.6E-12	62	202	2.8	Epstein-Barr virus infection
2.3E-13	70	234	2.7	Coronavirus disease-COVID-19
1.0E-07	41	143	2.6	Spinocerebellar ataxia
1.5E-09	56	203	2.5	Proteoglycans in cancer
3.1E-08	49	181	2.5	Herpes simplex virus 1 infection
8.4E-22	141	529	2.4	Pathways in cancer
4.7E-09	57	215	2.4	Chemical carcinogenesis-receptor activation
2.3E-13	88	332	2.4	Human papillomavirus infection
1.8E-08	53	200	2.4	Pathogenic Escherichia coli infection
1.0E-09	68	268	2.3	Parkinson disease
1.8E-12	95	387	2.2	Alzheimer disease

ShinyGO 0.85.1
Human

AFFX-HUMISGF3A
AFFX-HUMISGF3A
AFFX-HUMISGF3A
AFFX-HUMISGF3A
AFFX-HUMRGE
AFFX-HUMRGE
AFFX-r2-Ec-bioD-5_at

5617 IDs mapped to 2794 (50%) human genes.
Background (recommended) Submit

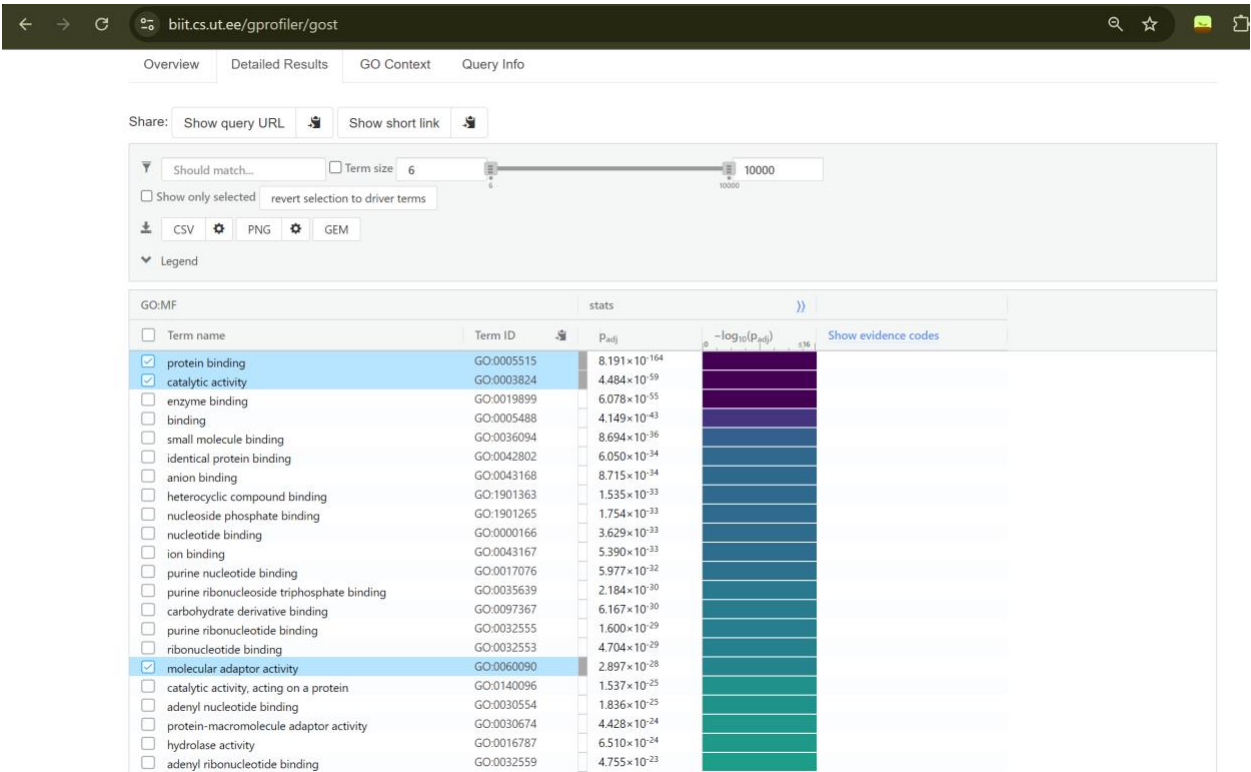
Pathway database: KEGG

FDR cutoff: 0,05 # pathways to show: 20

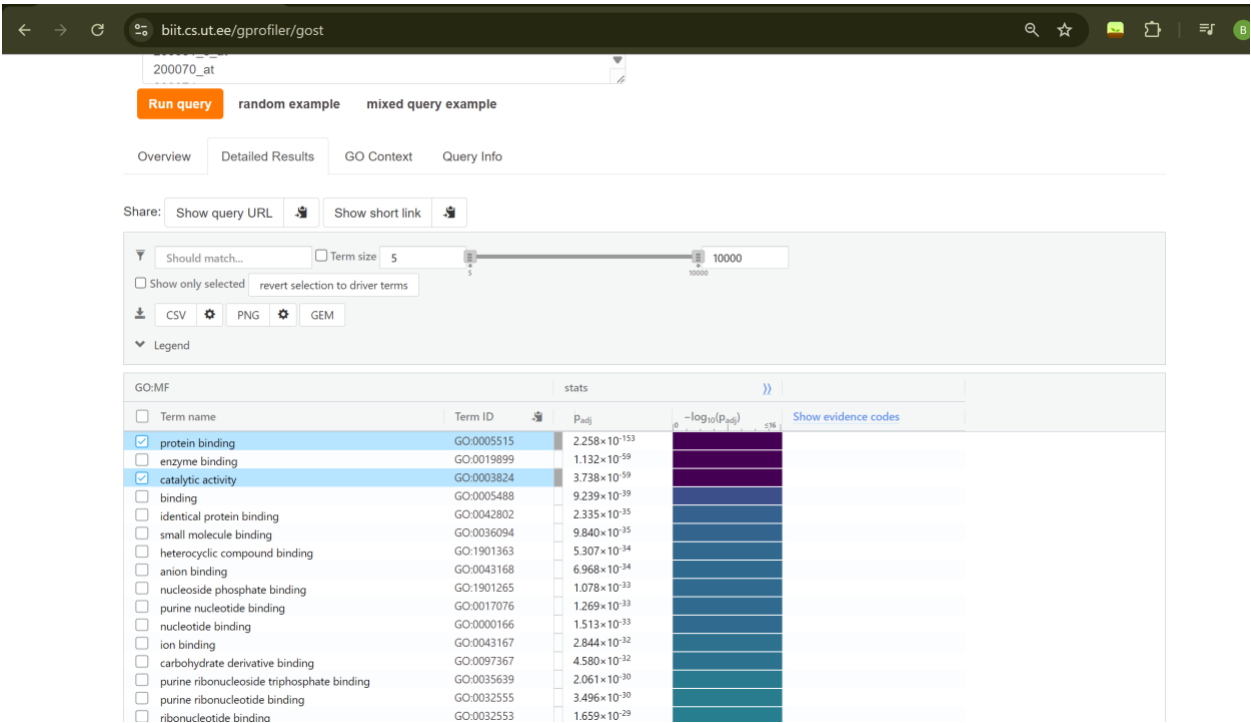
Pathway size: Min. 2 Max. 5000

Enrichment FDR	nGenes	Pathway Genes	Fold Enrichment	Pathways (click for details)
1.7E-09	23	46	4.5	Proteasome
1.1E-12	36	80	4.1	Drug metabolism-other enzymes
1.5E-09	29	69	3.8	Chemical carcinogenesis-DNA adducts
1.4E-08	36	109	3	HIIF-1 signalling pathway
1.2E-09	48	158	2.8	Cell cycle
6.2E-14	71	234	2.7	Coronavirus disease-COVID-19
4.9E-12	61	202	2.7	Epstein-Barr virus infection
1.7E-09	56	203	2.5	Proteoglycans in cancer
8.5E-24	145	529	2.5	Pathways in cancer
3.1E-14	90	332	2.5	Human papillomavirus infection
1.7E-09	58	215	2.4	Chemical carcinogenesis-receptor activation
1.6E-08	53	200	2.4	Pathogenic Escherichia coli infection
6.1E-08	52	202	2.3	Focal adhesion
8.3E-09	60	236	2.3	Ras signalling pathway
2.2E-09	67	268	2.3	Parkinson disease
2.0E-12	95	387	2.2	Alzheimer disease

When I manually erased about 10% of the bottom part, it fixed the issue. The image below shows the result of option F.



When I manually erased about 10% of the bottom part, it fixed the issue. The image below shows the result of option E. (I used E options result for more better results.)



(h) Discuss the enrichment analysis results: Considering the difference in the compared conditions (i.e. psoriasis versus normal), do you think the identified enriched terms are logical? Select two “biological process” or “KEGG pathway” terms that are common in the results of genes from (d) and (f). Select one term that only appears in the enrichment result of (d). Select one term that only appears in the enrichment result of (f). Search these four terms to see if they are related to psoriasis. Feel free to ask this to chatgpt. Discuss in detail.

I selected **protein binding and enzyme binding** as a common biological process. Yes, considering the comparison between psoriasis and normal skin conditions, the identified enrichment terms are reasonable. Psoriasis is a complex disease involving the upregulation of inflammatory pathways and the downregulation of normal skin barrier functions. Similarly, **enzyme binding biological pathways** are also expected to function improperly in this disease. Just one of them has **hydrolase activity** and other one doesn't have different pathway. Hydrolase activity in psoriasis, mainly from proteases, lipases, and phosphatases, contributes to the disease by amplifying immune responses and promoting keratinocyte hyperproliferation. It does not start the disease by itself but helps maintain and intensify the chronic inflammatory cycle.

i From the expression data, select 5 genes with lowest uncorrected p-values. Calculate Pearson correlation and Spearman correlation between each pair for the normal samples, and separately for the psoriasis samples. Are there any pairs that show significant correlation in one condition, and the significance is lost in the other condition? Below I gave lost correlation it means other have correlation

```

burak - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
+ - Source on Save
1 library(GEOquery)
2 TxData1 = getGEO("GSE52471")
3 TxData1 = getGEO("GSE52471",AnnotGPL = TRUE)
4 TT = TxData1[[1]]
5
6 a = exprs(TT)
7 b = pData(TT)
8 c = fData(TT)
9
10 group1_cols = grep("^Psoriasis", b$title)
11 group2_cols = grep("^Normal", b$title)
12
13
14 pval = NULL
15 for (i in 1:17440) {
16   pval[i] = t.test(a[,group1_cols],a[,group2_cols])$p.value
17 }
18
19 names(pval) = c$Gene symbol
20 pval = sort(pval)
21 head(names(pval))
22
23 length(which(pval<=0.05))
24 length(which(pval<=0.01))
25
26 bonferroni_pval = NULL
27 num = 0
28 for (i in pval) {
20:18 (Top Level)

```

Console Background Jobs

```

R - R 4.5.1 - ~/burak/
> for (i in 1:17440) {
+   pval[i] = t.test(a[,group1_cols],a[,group2_cols])$p.value
+ }
>
> names(pval) = c$Gene symbol
> pval = sort(pval)
> head(names(pval))
[1] "S100A12" "KYNNU" "CLDN8" "SPRR2C" "KYNNU" "IFI27"
>

```

```

17
18
19 names(pval) = c$Gene symbol
20 top5 = order(pval)[1:5]
21 #top 5 genes by lowest p-value
22 top5 = order(pval)[1:5]
23
24 #Iterate gene pairs
25 for(i in 1:4) for(j in (i+1):5) {
26   # Calculate p-values for Normal and Psoriasis groups (firs is pearson i'm doing)
27   p_norm = cor.test(a[top5[i], group2_cols], a[top5[j], group2_cols])$p.value
28   p_pso = cor.test(a[top5[i], group1_cols], a[top5[j], group1_cols])$p.value
29
30   # 3. Check if significance is lost (Significant in Normal -> Not significant in Psoriasis)
31   if(p_norm < 0.05 & p_pso > 0.05)
32     print(paste(c$Gene symbol[top5[i]], "and", c$Gene symbol[top5[j]], "lost correlation! pearson"))
33 }
34
35 # Now for spearman
36 for(i in 1:4) for(j in (i+1):5) {
37   # Calculate p-values for Normal and Psoriasis groups
38   p_norm = cor.test(a[top5[i], group2_cols], a[top5[j], group2_cols])$p.value
39   p_pso = cor.test(a[top5[i], group1_cols], a[top5[j], group1_cols])$p.value
40
41   # if significance is lost
42   if(p_norm < 0.05 & p_pso > 0.05)
43     print(paste(c$Gene symbol[top5[i]], "and", c$Gene symbol[top5[j]], "lost correlation! spearman"))
44 }
43:104 (Top Level)

```

```

[1] "S100A12 and KYNNU lost correlation! spearman"
[1] "KYNNU and SPRR2C lost correlation! spearman"
[1] "CLDN8 and SPRR2C lost correlation! spearman"
>
+ }
[1] "S100A12 and KYNNU lost correlation! pearson"
[1] "KYNNU and SPRR2C lost correlation! pearson"
[1] "CLDN8 and SPRR2C lost correlation! pearson"
>

```

(j) (must be hand-written!) What is the goal of this Homework? Also, explain what you learned in five bullet points.

The goal of this assignment was to learn the complete of transcriptomic data analysis, specifically managing datasets in R, performing statistical tests, and interpreting biological data to understand disease mechanisms.

Five bullet points

Data management, How to manipulate raw datasets

Statistical analysis, How to apply tests to data

Normalization, log transformation

Biological interpretation, How to perform enrichment and interpretation

Correlation Analysis, How to calculate Pearson and Spearman correlations to observe how gene-gene interactions change between conditions.

extras

The screenshot displays the RStudio environment with three main components: a script editor, a data viewer, and a console.

Script Editor: The script contains R code for loading data from GEO. The first version (lines 1-9) defines variables for gene expression data and sample/feature data. The second version (lines 1-10) adds the `b$title` command to the script.

```
1 library(GEOquery)
2 TxData1 = getGEO("GSE52471")
3 TxData1 = getGEO("GSE52471", AnnotGPL = TRUE)
4 TT = TxData1[[1]]
5
6 a = exprs(TT) # numerical data
7 b = pData(TT) # meta data on samples
8 c = fData(TT) # meta data on genes
9
10 b$title
```

Data Viewer: A table showing the first 12 rows of the `b` data frame. The columns are `data`, `row count`, `relation`, `relation.1`, `condition:ch1`, `kit:ch1`, `patient:ch1`, and `tissue:ch1`.

	data	row count	relation	relation.1	condition:ch1	kit:ch1	patient:ch1	tissue:ch1
1267nnn/GSM1...	17440	Reanalysis of: GSM300750		Psoriasis	Affy	PS-A	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300756		Psoriasis	Affy	PS-B	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300762		Psoriasis	Affy	PS-C	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300768		Psoriasis	Affy	PS-D	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300774		Psoriasis	Affy	PS-E	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300780		Psoriasis	Affy	PS-F	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300786		Psoriasis	Affy	PS-G	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300792		Psoriasis	Affy	PS-H	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300798		Psoriasis	Affy	PS-I	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300804		Psoriasis	Affy	PS-J	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300810		Psoriasis	Affy	PS-K	LS skin biopsy	
1267nnn/GSM1...	17440	Reanalysis of: GSM300815		Psoriasis	Affy	PS-L	LS skin biopsy	

Console: The console shows the output of the `view(a)`, `view(b)`, and `b$title` commands. The first two commands show the dimensions of the data frames. The `b$title` command displays a list of sample names.

```
R - R 4.5.1 - ~/burak/
GO:0071595//GO:0005737//GO:0005789//GO:0016021//GO:0005635//GO:0005635//GO:0031965
[ reached 'max' / getOption("max.print") -- omitted 17393 rows ]
> view(a)
> view(b)
> b$title
[1] "Psoriasis-Patient-A" "Psoriasis-Patient-B" "Psoriasis-Patient-C" "Psoriasis-Patient-D"
[5] "Psoriasis-Patient-E" "Psoriasis-Patient-F" "Psoriasis-Patient-G" "Psoriasis-Patient-H"
[9] "Psoriasis-Patient-I" "Psoriasis-Patient-J" "Psoriasis-Patient-K" "Psoriasis-Patient-L"
[13] "Psoriasis-Patient-M" "Psoriasis-Patient-N" "Psoriasis-Patient-O" "Normal-N_LJ_1"
[17] "Normal-N_LJ_2" "Normal-N_LJ_3" "Normal-N_LJ_4" "Normal-N_LJ_5"
[21] "Normal-N_LJ_6" "Normal-N_LJ_7" "Normal-N_LJ_8" "Normal-N_LJ_9"
[25] "Normal-N_LJ_10" "Discoid Lupus-L1" "Discoid Lupus-L2" "Discoid Lupus-L3"
[29] "Discoid Lupus-L4" "Discoid Lupus-L5" "Discoid Lupus-L8" "Discoid Lupus-L9"
[33] "Normal-N_AJ1" "Normal-N_AJ2" "Normal-N_AJ3" "Psoriasis-252"
[37] "Psoriasis-287" "Psoriasis-292"
```